APPLIED STATISTICAL METHODS

Assignment 1
Extensions of Analysis of Variance

Group Members

Anangam Saluja - 2024A2PS1308P

Adit Kothari - 2024A8PS0638P

Kushagra Agrawal - 2024A2PS1347P

Pulkit Dwivedi - 2024A2PS0342P

Rahul Bhat - 2023B4A31035P

Navaneetha G - 2023B4A71046P

Anushka Bhandari - 2023B4A70902P

TABLE OF CONTENTS

1 One-way ANOVA 3

1.1 Principle and Concept of ANOVA 3

2 Two-way ANOVA 4

2.1 Introduction to the Concept 4

2.2 Assumptions 4

2.3 Mathematical Model 4

2.4 Calculations 4

2.5 Hypothesis Testing 5

2.6 Steps / Methodology 5

2.7 Worked Example 6

2.8 Code 6

3 Three-way ANOVA 7

3.1 Introduction to the Concept 7

3.2 Assumptions 7

3.3 Mathematical Model 7

3.4 Calculations 8

3.5 Hypothesis Testing 8

3.6 Steps / Methodology 9

4 ANCOVA - Analysis of Covariance 10

4.1 Introduction to the Concept 10

4.2 Case Study / Example 10

4.3 Mathematical Model 10

4.4 Assumptions 11

If an assumption fails 11

4.5 Steps / Methodology 11

4.6 Calculations 11

4.7 Observed Values 12

4.8 Hypothesis Testing 13

Hypothesis 13

Test Statistic 13

4.9 Code 14

5 MANOVA - Multivariate Analysis of Variance 15

5.1 Introduction to the Concept 15

5.2 Assumptions 15

5.3 Mathematical Model 15

5.4 Steps / Methodology 16

5.5 Calculations 16

5.6 Hypothesis Testing 16

Hypothesis 16

Test Statistics..... 17
Decision Rule and Degrees of Freedom..... 17
5.7 Case Study / Example..... 17
Variables..... 17
Sample Sizes per Group (N = 395)..... 17
MANOVA Results..... 18
Follow-Up Univariate ANOVAs..... 18
5.8 Interpretation..... 18
5.9 Conclusion..... 19
5.10 Code..... 19
6 Conclusion..... 20
References..... 21

1 One-way ANOVA

1.1 Principle and Concept of ANOVA

ANOVA looks at the variance between groups relative to the variance within groups. The dependent variable (Y) must be metric, and the independent variable(s) (X), must be nominal. ANOVA is used to find out the main and interaction effects of categorical independent variables (called factors) on a dependent variable.

We know that one-way ANOVA helps to determine the effects of different levels of a single categorical variable on a continuous dependent variable, by analysing how the variation in the dependent variable splits into variance between groups and within groups. The test statistic which is Mean square due to treatment divided by Mean square error, tests if the means of the group are significantly different to not be a coincidence. If the group means don't differ significantly, we fail to reject the null hypothesis, hence the independent variable did not have a significant effect on the dependent variable. If they differ significantly, we reject the null hypothesis, and we run pairwise multiple comparison tests, to check which two groups differ.

However it is often not enough for more complex, real life problems where there may be more than one factor involved, more than one dependent variable or an external factor whose influence on the dependent variable may need to be controlled.

This report therefore provides a detailed description of three extensions of one-way ANOVA suited to these situations, namely Two-way ANOVA, ANCOVA and MANOVA, as well as a brief overview of Three-way ANOVA following the Two-way ANOVA section.

2 Two-way ANOVA

2.1 Introduction to the Concept

Two-way ANOVA is concerned with the comparison of the mean of a continuous response at all levels of two independent categorical factors. In this case, we have an independent factor A, which is a categorical variable with *a* levels representing the first group of treatment. Then we have another independent factor B, which is also a categorical variable with b levels representing the next group of treatments. We also have our dependent variable Y, which is a continuous variable, and it is the mean of Y we are studying in order to determine if factor A or B or their interaction has an effect on the response Y. One of the great benefits of two-way ANOVA is that it can be used in order to test for possible interaction between A and B.

2.2 Assumptions

Table 1. *Assumptions of two-way ANOVA and how to check them*

Assumption	How to check
Independence of observations	Study design (random assignment and no repeated measures)
Dependent variable is continuous; independent (factor) variables are categorical	Measurement level of the variables
Homogeneity of variance across all a×b treatment combinations (same as one-way ANOVA)	Levene's test on the residuals
Normality of residuals (same as one-way ANOVA)	Shapiro-Wilk test and a Q-Q plot
Additivity: effects of A and B combine additively except for the explicitly modelled interaction term	Interaction plot; significance test of the A×B interaction term

2.3 Mathematical Model

In the two-way ANOVA model, the response variable, Y_{ijk}, is modelled as the sum of the overall mean, the main effect of Factor A, the main effect of Factor B, the interaction effect of Factors A and B, and the random error. This can be expressed as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

where μ is the overall mean, α_i is the effect of the i-th level of Factor A, β_j is the effect of the j-th level of Factor B, $(\alpha\beta)_{ij}$ is the interaction effect of Factors A and B, and ϵ_{ijk} is the random error term. This model has the constraints $\Sigma\alpha_i = 0$, $\Sigma\beta_j = 0$, and the interaction effects have similar constraints so that the effects reflect deviations from the overall mean. When the interaction effect is significant, care must be taken when interpreting the main effects since the effect of one factor is conditional on the level of another factor, and simple effects are determined.

2.4 Calculations

Let a and b be the number of levels of A and B, respectively, n be the number of replicates per cell, so that N = abn. Let T be the grand total, the sum of all N scores. Similarly, let T_i and T_j be the sum of

scores for level i of factor A and level j of factor B respectively, and let T_ij be the sum of scores in one particular cell, i.e., for one combination of levels of A and B:

$$SS_A = (1/bn)\,\Sigma_i\, T_i^2 - T^2/N$$

$$SS_B = (1/an)\,\Sigma_j\, T_j^2 - T^2/N$$

$$SS_{AB} = (1/n)\,\Sigma_i\Sigma_j\, T_{ij}^2 - T^2/N - SS_A - SS_B$$

$$SS_{Total} = \Sigma_i\Sigma_j\Sigma_k\, Y_{ijk}^2 - T^2/N$$

$$SS_{Error} = SS_{Total} - SS_A - SS_B - SS_{AB}$$

Table 2. *Two-way ANOVA table*

Source	SS	df	MS	F	Tests
Factor A	SS_A	a−1	MS_A	F_A	Main effect of A
Factor B	SS_B	b−1	MS_B	F_B	Main effect of B
Interaction	SS_{AB}	(a−1)(b−1)	MS_{AB}	F_{AB}	A×B interaction
Error	SS_{Error}	ab(n−1)	MS_{Error}	—	—
Total	SS_{Total}	abn−1	—	—	—

2.5 Hypothesis Testing

Each effect – including interaction, along with other effects is being tested independently, having separate null and alternative hypotheses. In the case of interaction, hypotheses will look like:

H_0: $(\alpha\beta)_{ij} = 0$ for all i, j H_a: $(\alpha\beta)_{ij} \neq 0$ for at least one i, j

The test statistics is the ratio of the mean squares:

$F = MS_{effect} / MS_{error}$, where $MS_{effect} = SS_{effect} / df_{effect}$

Degrees of freedom for the variables are calculated by the (levels – 1) rule and are as follows: $df_A = a–1$, $df_B = b–1$. For interaction, they equal the product of two degrees of freedom: dfAB = (a–1)(b–1). Degrees of freedom for error are calculated as $df_{error} = ab(n-1)$, or N-ab.

If $p < \alpha$, H_0 is rejected, which means that the effect is statistically significant. Otherwise, H_0 is not rejected.

2.6 Steps / Methodology

1. Choose the factors A and B and collect data for each of the a×b cells, n replicates in each.
2. Compute the cell means, the marginal means for each factor, and the grand mean.
3. Calculate SS for all three effects (A, B, AB) and SS_{Error} from that and place all of these in the ANOVA table along with MS and F for each effect.
4. First look for interaction. If it's significant, there's your result to interpret. Make the interaction plot, not the main effects plot.
5. If the interaction is not significant, you may also interpret the two main effects separately.
6. Carry out post-hoc analysis for significant effect (Tukey HSD or Bonferroni)

2.7 Worked Example

Drawn from the same UCI Student Performance data used in the Case Study below: Factor A = weekly study time (<2h, >10h), Factor B = sex (F, M), response = final grade G3. For a hand-calculable illustration, a small balanced subsample of n = 3 students per cell (N = 12) is taken from the full dataset, so a = 2, b = 2, n = 3.

Table 3. *Worked example: final grades (G3) by study time and sex, n = 3 per cell*

	F	M	Row total (Ti.)
<2h	**14, 8, 6**	**14, 5, 10**	**T1. = 57**
>10h	**6, 16, 11**	**20, 12, 15**	**T2. = 80**
Col total (T.j)	**T.1 = 61**	**T.2 = 76**	**T = 137**

Step 1 — Grand correction term: $T^2/N = 137^2/12 = 1564.08$

Step 2 — Sums of squares:

$SS_A = (1/b_n)\Sigma T_i^2 - T^2/N = (57^2 + 80^2)/6 - 1564.08 = 44.08$

$SS_B = (1/a_n)\Sigma T_j^2 - T^2/N = (61^2 + 76^2)/6 - 1564.08 = 18.75$

$SS_{AB} = (1/n)\Sigma\Sigma T_{ij}^2 - T^2/N - SS_A - SS_B = 1641.00 - 1564.08 - 44.08 - 18.75 = 14.08$

$SS_{Total} = \Sigma Y^2 - T^2/N = 1799.00 - 1564.08 = 234.92$

$SS_{Error} = SS_{Total} - SS_A - SS_B - SS_{AB} = 158.00$

Step 3 — ANOVA table:

Table 4. *ANOVA table for the worked example*

Source	SS	df	MS	F
Study Group (A)	44.08	1	44.08	2.23
Sex (B)	18.75	1	18.75	0.95
Interaction (AB)	14.08	1	14.08	0.71
Error	158.00	8	19.75	—
Total	234.92	11	—	—

Conclusion: Fcrit(1, 8) at α = 0.05 is about 5.32. None of the F values (2.23, 0.95, 0.71) is greater than this critical value, so we can not reject H0 for study hours, sex and the interaction of these two factors, which means none of the factors are statistically significant. This is understandable because n = 3 per cell has very little power – and this is precisely why the full Case Study below uses the full dataset (N = 395) and not a tiny subsample, because in such a case study, both study hours and sex appeared to be statistically significant variables (see the ANOVA table in Case Study / Example).

2.8 Code

The code can be found here: [Python Code for Two-way ANOVA case study]

3 Three-way ANOVA

3.1 Introduction to the Concept

Three-way ANOVA is just two-way ANOVA, with one extra factor added. Instead of two categorical variables, Three-way ANOVA now has three factors – A, B and C – with a, b and c levels respectively. Three-way ANOVA checks whether any of these factors alone or together influence the variable Y. The main question stays the same: are the group means enough apart that random chance is unlikely to explain the difference? What changes is the number of interactions Three-way ANOVA must track. In two-way ANOVA there is one interaction, A×B. In Three-way ANOVA we have three two-way interactions: A×B, A×C and B×C in addition to a new three-way interaction, A×B×C. That three-way interaction asks whether the interaction between two factors changes when the third factor changes level. The feature that makes Three-way ANOVA more useful than running three Two-way ANOVAs also makes interpretation harder because an effect can now depend on two other factors at the same time instead of just one.

3.2 Assumptions

The assumptions carry over unchanged from one-way and two-way ANOVA, just extended to all a×b×c treatment combinations instead of fewer groups:

Table 5. *Assumptions of three-way ANOVA and how to check them*

Assumption	How to check
Independence of observations	Study design (random assignment and no repeated measures)
Dependent variable is continuous; independent (factor) variables A, B, C are categorical	Measurement level of the variables
Homogeneity of variance across all a×b×c treatment combinations (same as one-way/two-way ANOVA)	Levene's test on the residuals
Normality of residuals (same as one-way/two-way ANOVA)	Shapiro-Wilk test and a Q-Q plot
Additivity: effects of A, B, C combine additively except for the explicitly modelled interaction terms	Interaction plots; significance tests of the A×B, A×C, B×C, and A×B×C terms
Balanced design (equal n per cell)	Check cell sizes; if unbalanced, Type II/III sum-of-squares adjustments are required

3.3 Mathematical Model

Similar to two-way ANOVA, we can write the response as one additive model – the overall mean, the three main effects, the three two-way interactions, the three-way interaction, and whatever's left over as random error:

$$Y_{ijkl} = \mu + \alpha_i + \beta_j + \gamma_k + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} + (\alpha\beta\gamma)_{ijk} + \varepsilon_{ijkl}, \quad \varepsilon_{ijkl} \sim \text{iid } N(0, \sigma^2)$$

Table 6. *Terms in the three-way ANOVA model*

Variable	Explanation
Y_{ijkl}	dependent variable for the l-th subject at level i of A, j of B, k of C

Variable	Explanation
μ	overall (grand) mean
α_i, β_j, γ_k	main effects of the i-th level of A, j-th level of B, k-th level of C (constraints: $\Sigma\alpha i = \Sigma\beta j = \Sigma\gamma k = 0$)
$(\alpha\beta)_{ij}$, $(\alpha\gamma)_{ik}$, $(\beta\gamma)_{jk}$	two-way interaction effects between the respective factor pairs (zero-sum constraints apply)
$(\alpha\beta\gamma)_{ijk}$	three-way interaction effect among A, B, and C (zero-sum constraint applies)
ε_{ijkl}	random error associated with the observation

If the three-way term ends up becoming significant, that is the thing to focus on before trying to understand anything else – it shows that the two-way interactions are not acting the same way at every level of the third factor. So instead of looking at for example A×B by itself you would create a plot for each level of C and see how the results differ. Only if the three-way interaction isn't significant does it make sense to go back and look at the two-way interactions on their own, and only if those aren't significant either should you just interpret the three main effects independently.

3.4 Calculations

As in the two-way ANOVA section, it's easier to compute these using totals rather than means. With a, b, c as the number of levels of A, B and C, and n as the number of replicates per cell (so N = abcn, same as before), let T be the grand total – all N observations added up. T_i, T_j and T_k are the totals when you only sum within level i of A, level j of B, or level k of C. T_{ij}, T_{ik} and T_{jk} are the totals for each pair of factor levels, and T_{ijk} is the total for one specific cell, i.e. one combination of A, B and C:

$$SS_A = (1/bcn)\,\Sigma_i\, T_i^2 - T^2/N$$

$$SS_B = (1/acn)\,\Sigma_j\, T_j^2 - T^2/N$$

$$SS_C = (1/abn)\,\Sigma_k\, T_k^2 - T^2/N$$

$$SS_{AB} = (1/cn)\,\Sigma_i\Sigma_j\, T_{ij}^2 - T^2/N - SS_A - SS_B$$

$$SS_{AC} = (1/bn)\,\Sigma_i\Sigma_k\, T_{ik}^2 - T^2/N - SS_A - SS_C$$

$$SS_{BC} = (1/an)\,\Sigma_j\Sigma_k\, T_{jk}^2 - T^2/N - SS_B - SS_C$$

$$SS_{ABC} = (1/n)\,\Sigma_i\Sigma_j\Sigma_k\, T_{ijk}^2 - T^2/N - SS_A - SS_B - SS_C - SS_{AB} - SS_{AC} - SS_{BC}$$

$$SS_{Total} = \Sigma_i\Sigma_j\Sigma_k\Sigma_l\, Y_{ijkl}^2 - T^2/N$$

$$SS_{Error} = SS_{Total} - SS_A - SS_B - SS_C - SS_{AB} - SS_{AC} - SS_{BC} - SS_{ABC}$$

3.5 Hypothesis Testing

Each effect – every main effect and every interaction – gets tested separately with its own null and alternative hypothesis. For the three-way interaction it looks like this:

$$H_0: (\alpha\beta\gamma)_{ijk} = 0 \text{ for all i, j, k} \qquad H_a: \text{at least one } (\alpha\beta\gamma)_{ijk} \neq 0$$

The test statistic is the usual ratio of mean squares:

$$F = MS_{effect} / MS_{Error}, \text{ where } MS_{effect} = SS_{effect} / df_{effect}$$

The degrees of freedom follow the same (levels − 1) rule as before: $df_A = a-1$, $df_B = b-1$, $df_C = c-1$. For a two-way interaction it's just the product of the corresponding pair – e.g. $df_{AB} = (a-1)(b-1)$ – and for the three-way interaction, $df_{ABC} = (a-1)(b-1)(c-1)$. The error degrees of freedom are $df_{Error} = abc(n-1)$, which is the same thing as $N - abc$.

If $p < \alpha$ we reject H_0, which means that effect actually matters. If $p \geq \alpha$ we fail to reject H_0 – not enough evidence that it does.

3.6 Steps / Methodology

1. Pick out the three factors A, B and C, and collect data for every one of the a×b×c cells, with n replicates in each.
2. Work out the cell means, the marginal means for each factor, and the overall grand mean.
3. Use those to get SS for all seven effects plus SS_{Error}, then put it all into an ANOVA table along with MS and F for each one.
4. Check the three-way interaction first. If it's significant, that's the one to interpret – plot the two-way interaction separately at each level of the third factor rather than looking at main effects on their own.
5. If it's not significant, move down to the two-way interactions. If none of those are significant either, it's fine to just interpret the three main effects directly.
6. Whatever comes out significant, follow up with a post-hoc test (Tukey HSD or Bonferroni) to see exactly which levels are actually different from each other.

4 ANCOVA - Analysis of Covariance

4.1 Introduction to the Concept

ANCOVA compares the group means of different groups after statistically controlling the influence of one or more continuous variables called covariates. We have an independent factor, which is a categorical variable representing the different groups e.g., conditions or treatments administered. Then we have a continuous predictor variable X, called the covariate, measured before the treatment, whose influence on the dependent variable is to be controlled for. Finally, Y is the dependent variable whose adjusted group means are compared to determine whether the treatments have a significant effect. It is a combination of one-way ANOVA with regression, that helps to answer whether the group means would differ if they all started with the same value of the covariate.

We shall proceed to further explain ANCOVA along with corresponding steps in a case study for easier understanding.

4.2 Case Study / Example

The data set used is the UCI Student Performance dataset (Cortez & Silva, 2008)

We analyse a particular course, here mathematics.

Objective: To determine whether weekly study time affects the final mathematics grade, once each student's first-period grade is taken into account.

4.3 Mathematical Model

$$Y_{ij} = \mu + \tau_i + \beta\,(X_{ij} - \overline{X}) + \epsilon_{ij}; \qquad \epsilon_{ij} \sim iid\ N(0,\ \sigma^2)$$

Table 7. *Terms in the ANCOVA model and their meaning in the case study*

Variable	General explanation	Case Study Usage (Course : Mathematics)
τ_i	effect of the i-th treatment/group ($constraint:\ \Sigma\tau_i = 0$ meaning the group effects sum to 0)	Relative effect of weekly studying time for i= A, B, C, D Group A : < 2 hours, B : 2 to 5 hours, C : 5 to 10 hours, D : >10 hours
Y_{ij}	dependent variable for the jth observation in group i	Final score of student j in group i (0-20)
μ	overall mean of adjusted final test scores	Mean of all adjusted final test scores
X_{ij}	value of the covariate for the jth observation of ith group	Initial score of student j in group i (0-20)
β	regression coefficient (the effect of the covariate)	Regression coefficient of the effect of the initial test score in final score
ϵ_{ij}	Random error	Noise
$\overline{X}$	Covariate mean	Mean of all initial scores

4.4 Assumptions

Table 8. *Assumptions of ANCOVA and how to check them*

Assumption	How to check
Independence of observations	Study design (random assignment, no repeated measures)
Dependent variable should be continuous , Independent (factor/group) variable must be categorical	Measurement level of the variables
Covariate not affected by the treatments and measured reliably.	Covariate should be measured before treatment, and using a reliable instrument
Linearity of relationship between covariate and dependent variable for each group.	Scatterplot of Y against X for each group
Homogeneity of the regression slopes: similar relationship between covariate and DV across groups	Check the regression coefficients of Y and X within all groups, they must be approximately equal.
Homogeneity of variance (same as ANOVA)	Run the Levene's test on the residuals
Normality of residuals (same as ANOVA)	Run a Shapiro-Wilk test and a Q-Q plot

If an assumption fails

When regression slopes are different within groups, standard ANCOVA is invalid, switch to a multiple regression model with additional terms for incorporating interaction effects.

When variances are unequal and group sizes differ, confirm the result with robust (HC3) standard errors.

All of these tests pass, in the dataset used in the case study. They are elaborated more on in the code.

4.5 Steps / Methodology

1. Select n * k subjects (students) and randomly assign them into groups of n for k different treatments (weekly study time).
2. First, record Covariate data (X) (initial score) and then administer the respective treatments.
3. Record Dependent Data (Y) (final test score)
4. Conduct tests to ensure the assumptions are satisfied.
5. Compute adjusted SS between and within groups.
6. Compute the test statistic and conduct hypothesis testing.
7. If the null hypothesis is rejected, calculate and compare the adjusted means, and run post hoc pairwise comparison tests, such as Bonferroni or Sidak adjustments.

4.6 Calculations

Table 9. *Sums of squares and sums of products used in ANCOVA*

	Between groups	Within Groups
Sum of Squares for Y (dependent variable)	$SS_{bg} = \frac{\sum^{k}(\sum^{n} Y)^{2}}{n} - \frac{(\sum^{k}\sum^{n} Y)^{2}}{kn}$	$SS_{wg} = \sum^{k}\sum^{n} Y^{2} - \frac{\sum^{k}(\sum^{n} Y)^{2}}{n}$

Sum of Squares for X (covariate)	$SS_{bg(x)} = \frac{\sum^{k}(\sum^{n} X)^{2}}{n} - \frac{(\sum^{k}\sum^{n} X)^{2}}{kn}$	$SS_{wg(x)} = \sum^{k}\sum^{n} X^{2} - \frac{\sum^{k}(\sum^{n} X)^{2}}{n}$
Sum of Products	$SP_{bg} = \frac{\sum^{k}(\sum^{n} Y)(\sum^{n} X)}{n} - \frac{(\sum^{k}\sum^{n} Y)(\sum^{k}\sum^{n} X)}{kn}$	$SP_{wg} = \sum^{k}\sum^{n} XY - \frac{\sum^{k}(\sum^{n} X)(\sum^{n} Y)}{n}$

$k = number\ of\ groups,\ n = number\ of\ subjects\ per\ group$

$SS'_{bg} = Adjusted\ between\ group\ variation$

$SS'_{bg} = original\ between\ group\ variation\ -\ variation\ influenced\ by\ covariate$

$SS'_{bg} = SS_{bg} - [\frac{(SP_{bg} + SP_{wg})^{2}}{SS_{bg(x)} + SS_{wg(x)}} - \frac{(SP_{wg})^{2}}{SS_{wg(x)}}]$

$SS'_{wg} = Adjusted\ within\ group\ variation$

$SS'_{wg} = original\ within\ group\ variation\ -\ variation\ influenced\ by\ covariate$

$SS'_{wg} = SS_{wg} - \frac{(SP_{wg})^{2}}{SS_{wg(x)}}$

Note: these calculations assume equal n per group, i.e. N = kn.

$Adjusted\ group\ means: \overline{Y}'_{i} = \overline{Y}_{i} - \beta(\overline{X}_{i} - \overline{X})$

$Mean\ of\ adjusted\ group\ means: \mu = \frac{1}{k}\sum_{i=1}^{k} \overline{Y}'_{i}$

$Effect\ of\ the\ i^{th}\ treatment/group: \tau = \overline{Y}'_{i} - \mu$

$Regression\ coefficient\ estimated\ using\ pooled\ within\ group\ slope: \beta = \frac{SP_{wg}}{SS_{wg(x)}}$

4.7 Observed Values

Table 10. *Observed values: group means, adjusted means and group effects (N = 357)*

Study time (group)	**n_i**	**$\overline{X}_i$: G1**	**Raw means : $\overline{Y}_i$**	**Adjusted means : $\overline{Y}'_i$**	**Effect of the group : τ_i**
<2 h	92	11.03	11.47	11.68	0.07
2-5 h	182	10.90	11.07	11.40	-0.21
5-10 h	59	12.39	12.56	11.57	-0.04
>10 h	24	12.25	12.67	11.80	+0.19
Total	N = 357	$\overline{X}$ = 11.27	$\overline{Y}$ = 11.52	$\mu = 11.61$	$\sum\tau_i = 0$

Common slope of G3 on G1 for all groups : $\beta = 3192.99/3610.45 = 0.884$

Variation in final scores within groups explained by the pre-test scores $= \beta \times SP_{wg} = 2823.79$

Adjusted between-group variation $= SS'_{bg} = 133.06 - [\frac{(127.73 + 3192.99)^{2}}{(127.73 + 3610.45)} - 2823.79] = 6.99$

Adjusted within-group variation in final test scores = $SS'_{wg} = 3575.99 - 2823.79 = 752.20$

4.8 Hypothesis Testing

Hypothesis

$$H_0: \tau_1 = \tau_2 = \ldots = \tau_k = 0 \text{ or } H_0: \mu_1^{adj} = \mu_2^{adj} = \ldots = \mu_k^{adj}$$

$$H_a: \textit{at least one adjusted group mean differs}$$

Test Statistic

$$F = \frac{\textit{adjusted between group variation}}{\textit{adjusted error variation}} = \frac{MSTR_{bg,\ adjusted}}{MSE_{wg,\ adjusted}} = \frac{SS_{bg,adjusted}/(k-1)}{SS_{wg,adjusted}/(N-k-1)}$$

$$p < \alpha \Rightarrow \textit{Reject } H_0 \Rightarrow \textit{atleast one treatment has significant effect on the subjects after controlling } X$$

$$p \geq \alpha \Rightarrow \textit{Fail to Reject } H_0 \Rightarrow \textit{no significant evidence of a difference between adjusted means}$$

Degree of freedom (between groups) is k-1, like in ANOVA.

Degree of freedom (within groups, i.e., error) is N-k -1, since we estimate an additional slope β

Table 11. *One-way ANOVA and ANCOVA results for the case study*

	ANOVA	**ANCOVA**
Hypothesis Testing	$H_0: \mu_A = \mu_B = \mu_C = \mu_D$ $H_a:$ *at least one group mean differ*	$H_0: \tau_A = \tau_B = \tau_C = \tau_D = 0$, or $H_0: \mu_A^{adj} = \mu_B^{adj} = \mu_C^{adj} = \mu_D^{adj}$ $H_a:$ *at least one adjusted group mean differs*
Between Group MS	$MSTR_{bg} = 44.35$	$MSTR'_{bg} = 2.33$
MSE (within group)	$MSE_{wg} = 10.13$	$MSE'_{wg} = 2.14$
Test Statistic	$F_{ANOVA} = 4.38$ $p_{ANOVA} = 0.005$	$F_{ANCOVA} = 1.09$ $p_{ANCOVA} = 0.353$
Result ($taking\ \alpha = 5\%$)	$p < 0.05 \rightarrow$ Rejected null hypothesis	$p \geq 0.05 \rightarrow$ Failed to reject null hypothesis
Conclusion	The time duration of studying had a significant effect on the final grades of the students.	After controlling for the initial test scores (understood as starting points of the students in a race) we conclude that there is no significant evidence suggesting that the duration of studying has an effect on the grades of the students.

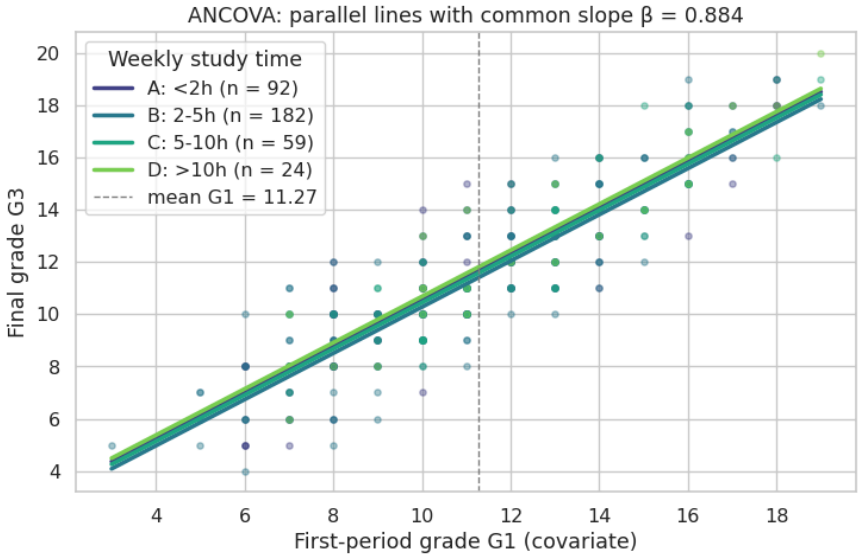


Figure 1. *ANCOVA fit: final grade (G3) against first-period grade (G1), with parallel regression lines for each study-time group (common slope β = 0.884)*

This plot shows each student plotted by their first period grade vs their final grade (the dots) by each group. The points are rising steadily from left to right confirming a strong positive correlation as well as linear relationship between G1 and G3. The four regression lines are parallel with the common slope $\beta = 0.884$ Since the vertical gap between the regression lines, that represents the pure effect of the study time on the final grades, is almost negligible, it shows that students with the same first period grade achieve almost the same final period grade regardless of their study time.
The dashed line indicates the average G1 (11.27) and intersects the regression lines at the adjusted means for each group, all of which lie within 0.4 points of each other.

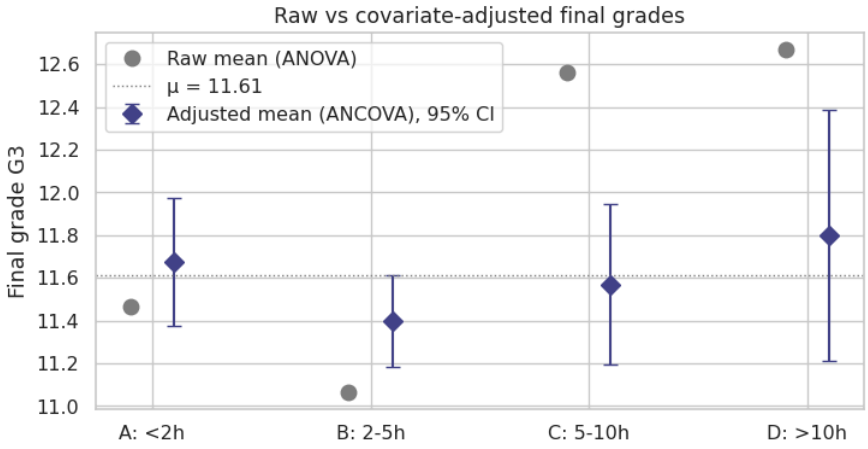


Figure 2. *Raw and covariate-adjusted mean final grades by weekly study time, with 95% confidence intervals*

The raw means suggest that students who study more score higher on the final test. However, after adjusting for the effects of G1, the last two study time groups are adjusted down to their expected values at the average G1 (11.27), while the first two groups are adjusted upward. The adjusted means all lie within 0.4 points of each other and are close to the average of 11.61 . Their confidence intervals also overlap which matches our deduction from the hypothesis testing.
Overall it supports the null hypothesis that study time makes no significant difference to the final grade.

4.9 Code

The code can be found here: [Python Code for ANCOVA case study]

5 MANOVA - Multivariate Analysis of Variance

5.1 Introduction to the Concept

Multivariate Analysis of Variance (MANOVA) is a method that looks at how one or more categorical independent variables influence several dependent variables at once. While ANOVA checks one response variable at a time, MANOVA checks a set of variables together and asks whether the groups differ in their total multivariate response.

In real-world studies, dependent variables are linked in most cases. For example, in a study, students' mathematics, physics, and chemistry scores may be correlated. Running ANOVAs for each subject raises the chance of finding a statistically significant result just by luck because of the build-up of Type I error. MANOVA solves this problem by testing all the variables at once and considering the correlations among them.

5.2 Assumptions

1. Observations should be independent.
2. Data is collected through random sampling.
3. DVs have multivariate normality within groups.
4. Covariance between DVs should be similar across groups.
5. The dependent variables are linearly related to each other within each group.

5.3 Mathematical Model

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}; \quad \epsilon_{ij} \sim iid\, N_p(0, \Sigma)$$

where,

Σ is the $p \times p$ covariance matrix

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \cdots & \sigma_p^2 \end{bmatrix}$$

$Y_{ij} = [Y_{ij1}\ Y_{ij2}\ Y_{ij3} \ldots . \ Y_{ijp}\ ]^T$: response vector for j-th observation in i-th group

$\mu = [\mu_1\ \mu_2\ \mu_3 \ldots\ \mu_p]^T$: overall population mean vector

$\tau_i = [\tau_{i1}\ \tau_{i2}\ \tau_{i3} \ldots . \ \tau_{ip}\ ]^T$ **:** effect of the i-th group (represents deviation of group mean vector from overall population mean vector)

$\epsilon_{ij} = [\epsilon_{ij1}\ \epsilon_{ij2}\ \epsilon_{ij3} \ldots . \ \epsilon_{ijp}\ ]^T$ **:** random error vector associated with j-th observation in i-th group

5.4 Steps / Methodology

1. Identify groups/ treatments and record multiple dependent variables for each observation.
2. Calculate mean vector for each group and overall mean vector.
3. Compute within-group variation and between-group variation.
4. Use MANOVA test statistics to conduct hypothesis testing.
5. If the null hypothesis is rejected, perform univariate ANOVAs and post hoc comparisons to find DVs and groups contributing to the difference.

5.5 Calculations

1. Group mean vector and overall mean vector.

$$\bar{Y}_i = \frac{1}{n_i}\sum_{j=1}^{n_i} Y_{ij} \qquad \bar{Y} = \frac{1}{N}\sum_{i=1}^{g}\sum_{j=1}^{n_i} Y_{ij}$$

n_i : number of observations in group i

N : total observations

$\bar{Y}_i$: mean vector of group i

$\bar{Y}$: overall mean vector

2. Within-group SSCP matrix and between-group SSCP matrix.

Table 12. *Between-group (H) and within-group (E) SSCP matrices*

$H = \sum_{i=1}^{g} n_i(\bar{Y}_i - \bar{Y})(\bar{Y}_i - \bar{Y})^T$	$E = \sum_{i=1}^{g}\sum_{j=1}^{n_i}(Y_{ij} - \bar{Y}_i)(Y_{ij} - \bar{Y}_i)^T$
H : between-group variation matrix g: total number of groups n_i : number of observations in group i $\bar{Y}_i$: mean vector of group i $\bar{Y}$: overall mean vector	E: within-group variation matrix Y_{ij}: response vector of observation j in group i

Total-group variation matrix: $T = H + E$

3. Eigenvalues

Solve $|H - \lambda E| = 0$ to get eigenvalues $\lambda_1, \lambda_2, ..., \lambda_s$, where $s = \min(p, g - 1)$

5.6 Hypothesis Testing

Hypothesis

Table 13. *Hypotheses for the MANOVA test*

$H_0: \mu_1 = \mu_2 = \cdots = \mu_g$ H_1: *At least one group mean vector differs*	$H_0: \tau_1 = \tau_2 = \cdots = \tau_g = 0$ H_1: *at least one* $\tau_i \neq 0$

Test Statistics

Table 14. *MANOVA test statistics*

Wilk's Lambda	$\Lambda = \prod_{k=1}^{s} \frac{1}{1+\lambda_k}$ or $\Lambda = \frac{\lvert E \rvert}{\lvert E+H \rvert}$
Pillai's Trace	$V = \sum_{k=1}^{s} \frac{\lambda_k}{1+\lambda_k}$
Hotelling–Lawley Trace	$U = \sum_{k=1}^{s} \lambda_k$
Roy's Greatest Root	$\Theta = \max(\lambda_1, \ldots, \lambda_s)$

Decision Rule and Degrees of Freedom

p-value $\leq \alpha \rightarrow$ Reject H_0

p-value $> \alpha \rightarrow$ Fail to reject H_0

Between-group d.o.f = g -1

Within-group d.o.f = N - g

5.7 Case Study / Example

In our example, we want to see whether a student's weekly study time has a significant effect on academic performance. The performance is measured across three related grade outcomes.

MANOVA is used here instead of running three separate ANOVAs because of the three grade outcomes being highly intercorrelated, testing them one by one, i.e using ANOVA would increase the probability of a Type I error.

In this example, we have considered α to be 0.05.

Variables

Table 15. *Variables in the MANOVA case study*

	Variable	**Description**	**Type**
Independent variable (factor)	study time	Weekly study time (1 = <2h, 2 = 2–5h, 3 = 5–10h, 4 = >10h)	Categorical, 4 levels
Dependent variables	G1,G2,G3	First-period, second-period, and final grades (0–20 scale)	Continuous

Sample Sizes per Group (N = 395)

Table 16. *Sample sizes per study-time group*

Study Time Group	n
<2h	105
2–5h	198
5–10h	65
>10h	27

MANOVA Results

Effect of study time on the combined outcome vector (G1, G2, G3):

Table 17. *MANOVA results: effect of study time on (G1, G2, G3)*

Test Statistic	Value	Num DF	Den DF	F	p-value
Wilks' Lambda	0.952	9	946.87	2.158	0.023
Pillai's Trace	0.049	9	1173.00	2.148	0.023
Hotelling–Lawley Trace	0.050	9	608.11	2.166	0.023
Roy's Greatest Root	0.038	3	391.00	4.993	0.002

All four test statistics are significant at $\alpha = 0.05$, as p-value ≤ 0.05.

Follow-Up Univariate ANOVAs

Table 18. *Follow-up univariate ANOVAs*

Dependent Variable	Sum of Squares (studytime)	df	F	p-value	Significant (α=0.05)?
G1	146.39	3	4.549	0.004	Yes
G2	116.98	3	2.794	0.040	Yes
G3	108.20	3	1.728	0.161	No

5.8 Interpretation

MANOVA results show that weekly study time has a significant effect on academic performance. The value of the test statistic is 0.049, with an F statistic of 2.15 and a p-value of 0.023 using Pillai's Trace, which indicates that the overall effect across all the grades is significant.

Considering each grade separately through ANOVAs, effect is mainly present in G1 and G2. However, G3 is not significant.

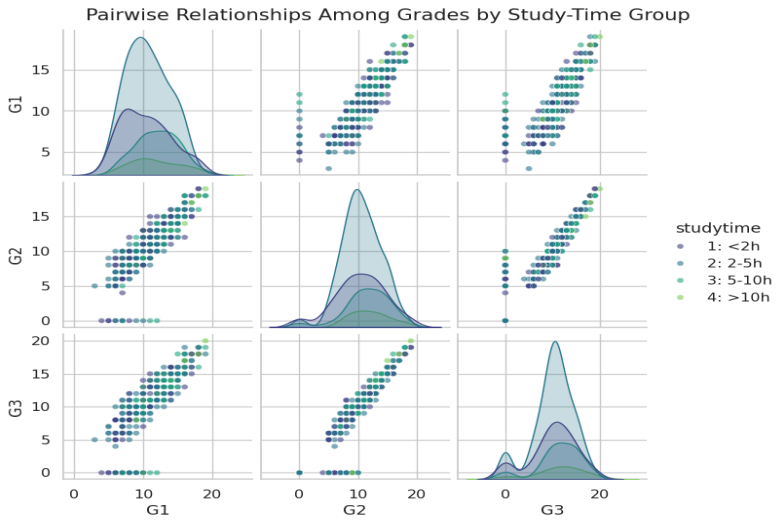


Figure 3. *Pairwise relationships among G1, G2 and G3 by study-time group*

Since Box's M indicated unequal covariance matrices across groups, Pillai's Trace is reported as the primary result and should be taken as the main test statistic.

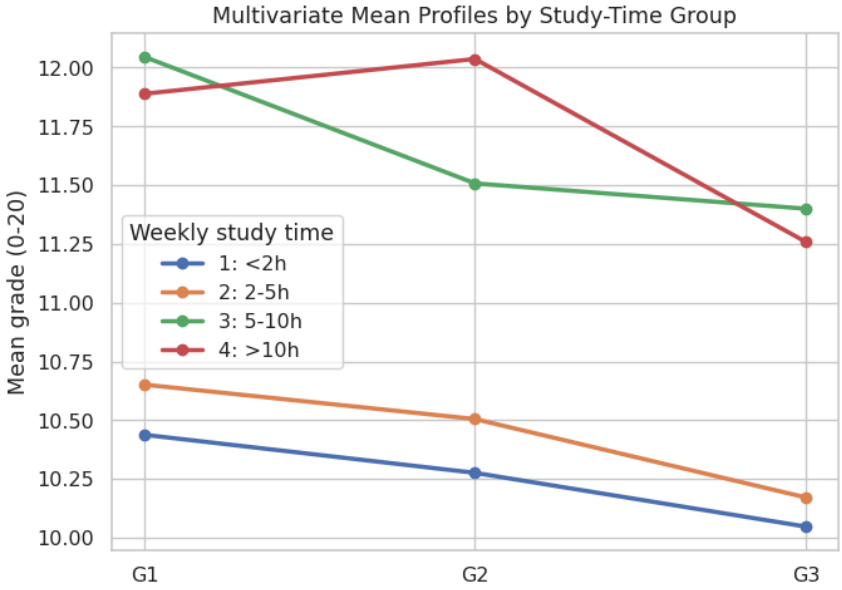


Figure 4. *Mean grade profiles (G1, G2, G3) by weekly study time*

5.9 Conclusion

Weekly study time is linked to students' academic performance when all grades are considered together. Pillai's Trace is 0.05 meaning weekly study time accounts for only a small part of the differences in the grades. Weekly study time has the link to early-term grades(G1 and G2) and the link becomes weaker by the final grade(G3).

5.10 Code

The code can be found here: manova_analysis.ipynb

6 Conclusion

One-way ANOVA answers a single question: do the means of one dependent variable differ across the levels of one factor / independent variable? The extensions discussed in this report and their differences are summarized here:

Applying the methods to the same student-performance data shows how the choice of method determines the conclusion:

- **Two-way ANOVA:** study time and sex both had small but significant main effects on the final grade ($p = .001$ and $p = .008$), with no interaction ($p = .792$).
- **ANCOVA:** the apparent effect of study time on the final mathematics grade ($F = 4.38$, $p = .005$) disappeared once the first-period grade was controlled for ($F = 1.09$, $p = .353$). Students who studied more were already performing better at the start.
- **MANOVA:** study time had a significant joint effect on G1, G2 and G3 (Pillai's $V = 0.049$, $p = .023$), even though G3 on its own was not significant, showing that a multivariate test can detect effects that single-outcome analyses miss.

Three-way ANOVA extends the factorial idea to three factors and their interactions, the cost of larger samples. Choosing the appropriate extension thus depends on the number of factors, the number of outcomes and whether a covariate must be controlled for. Since our data was observational, all effects found are associations rather than causal effects.

References

Cortez, P. (2008). *Student performance* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T

Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school student performance. In A. Brito & J. Teixeira (Eds.), *Proceedings of the 5th Future Business Technology Conference (FUBUTEC 2008)* (pp. 5–12). EUROSIS.

Field, A. (2018). *Discovering statistics using IBM SPSS Statistics* (5th ed.). SAGE.

Huitema, B. E. (2011). *The analysis of covariance and alternatives* (2nd ed.). Wiley.

Johnson, R. A., & Wichern, D. W. (2007). *Applied multivariate statistical analysis* (6th ed.). Pearson Prentice Hall.

Keller, B., & Marchev, D. (2022). Analysis of covariance: Univariate and multivariate approaches. In R. J. Tierney, F. Rizvi, & K. Ercikan (Eds.), *International encyclopedia of education* (4th ed.). Elsevier.

Maxwell, S. E., Delaney, H. D., & Kelley, K. (2018). *Designing experiments and analyzing data: A model comparison perspective* (3rd ed.). Routledge.

Montgomery, D. C. (2017). *Design and analysis of experiments* (9th ed.). Wiley.

Rencher, A. C., & Christensen, W. F. (2012). *Methods of multivariate analysis* (3rd ed.). Wiley.

Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with Python. In *Proceedings of the 9th Python in Science Conference* (pp. 92–96).

Tabachnick, B. G., & Fidell, L. S. (2019). *Using multivariate statistics* (7th ed.). Pearson.

Virtanen, P., Gommers, R., Oliphant, T. E., et al. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods, 17*, 261–272.